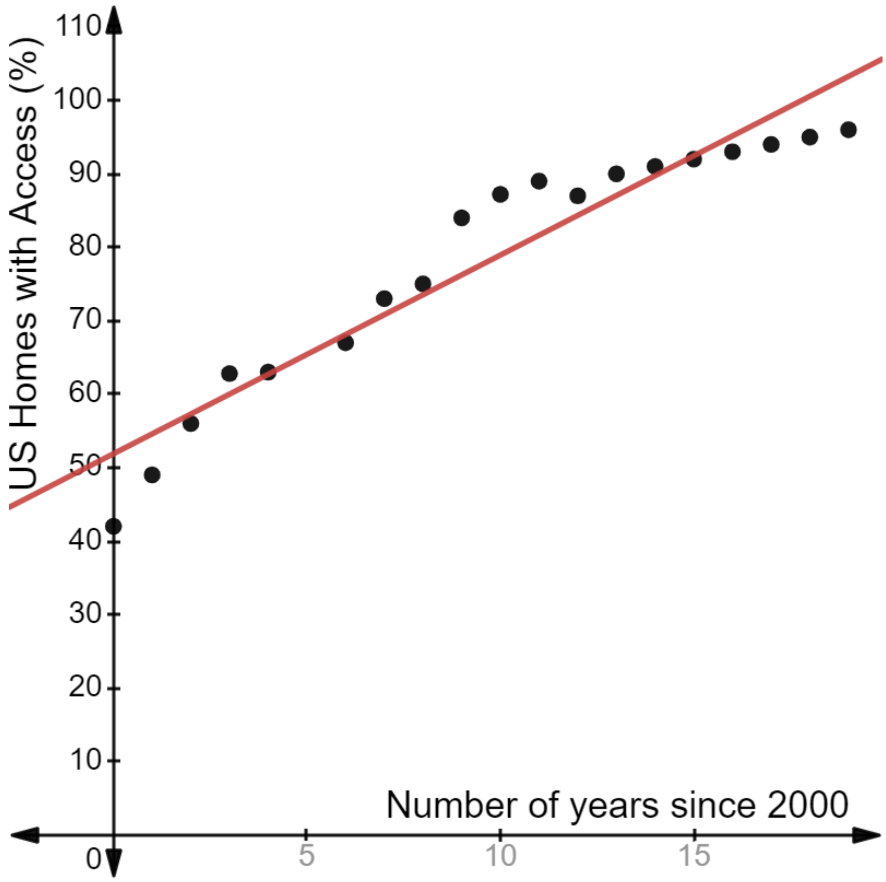
Grade 8
Unit 11 Assessment

Name____________________
Date_________________
Period__________

1. A scatterplot of the percentage of US households with access to cell phones since the year 2000 is shown.



a. Draw a line on the scatter plot that fits the data.

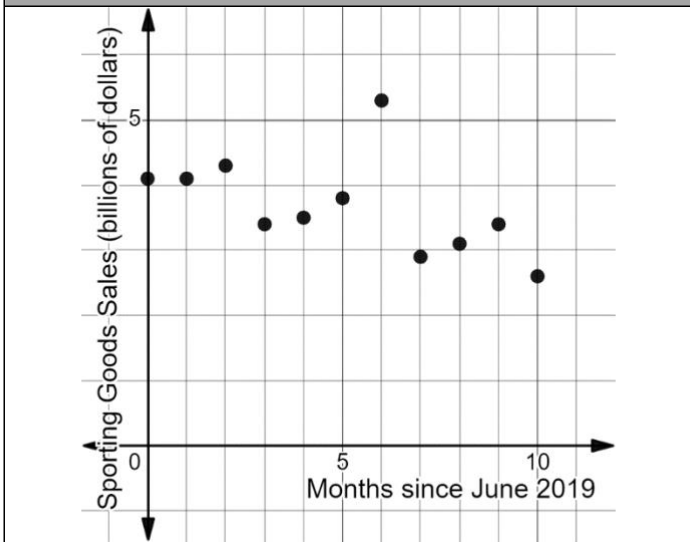
b. Complete the statement.

The line of fit shows

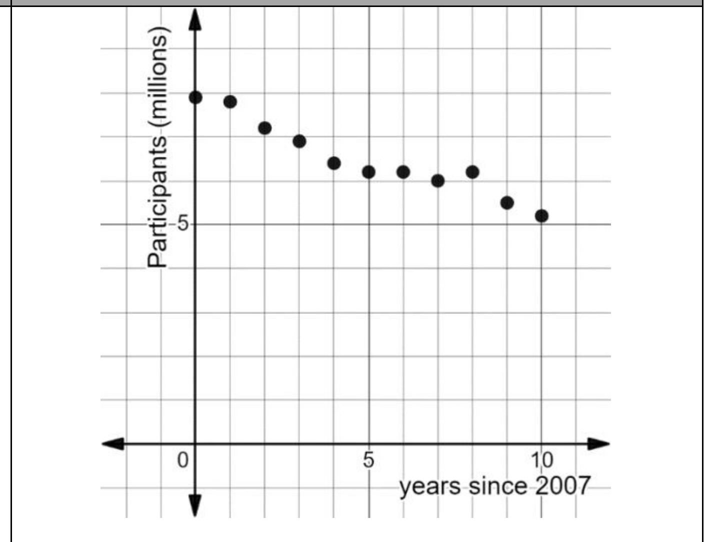
- ○ a closer analysis from each point to the other in the dataset.
- ○ that the independent variable is the percentage of US households with access to cell phones.
- ○ the overall relationship between the number of years since 2000 and the percentage of US households with access to cell phones.
- ○ that there is no association between the number of years since 2000 and the percentage of US households with access to cell phones.

2. Six scatter plots are shown.

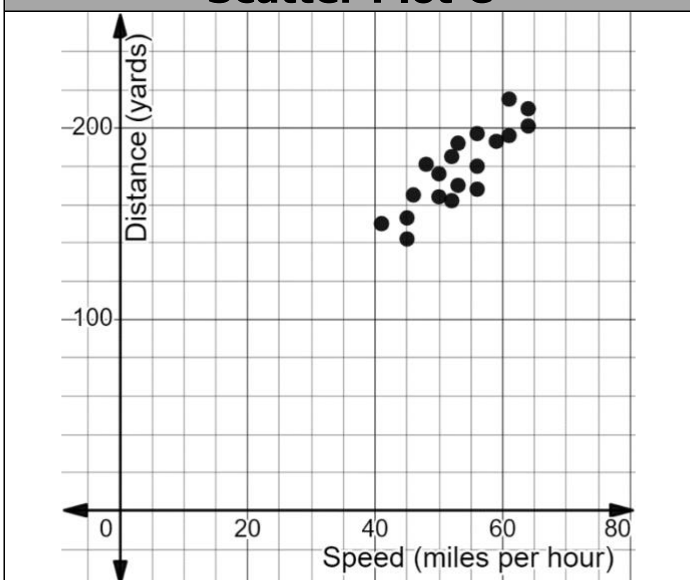
Scatter Plot A



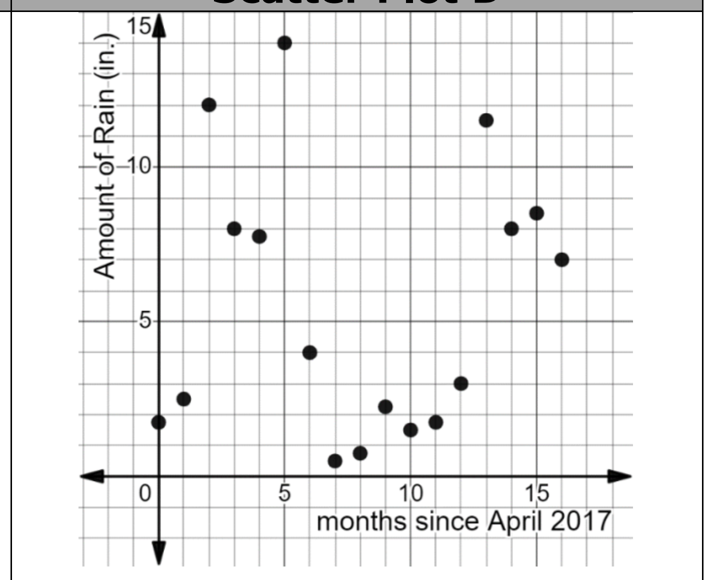
Scatter Plot B



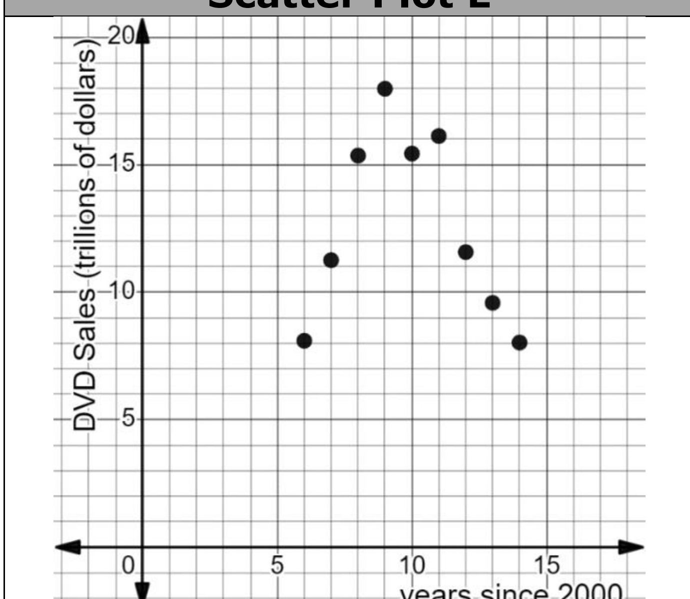
Scatter Plot C



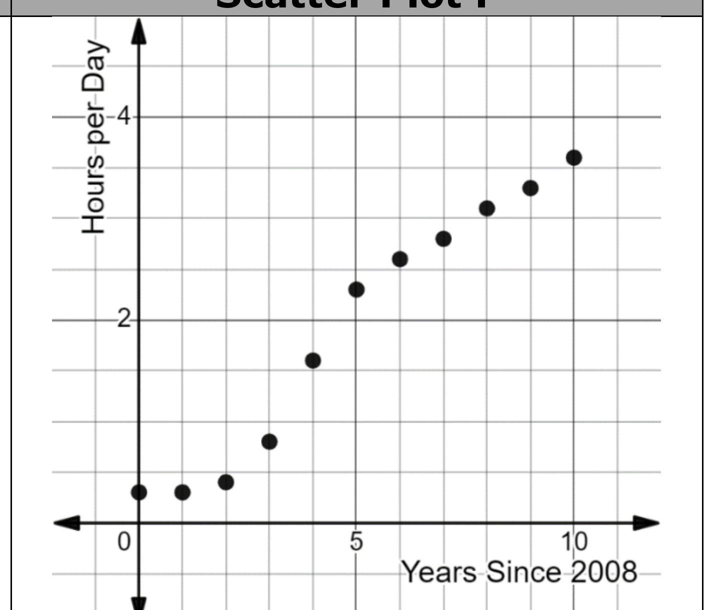
Scatter Plot D



Scatter Plot E



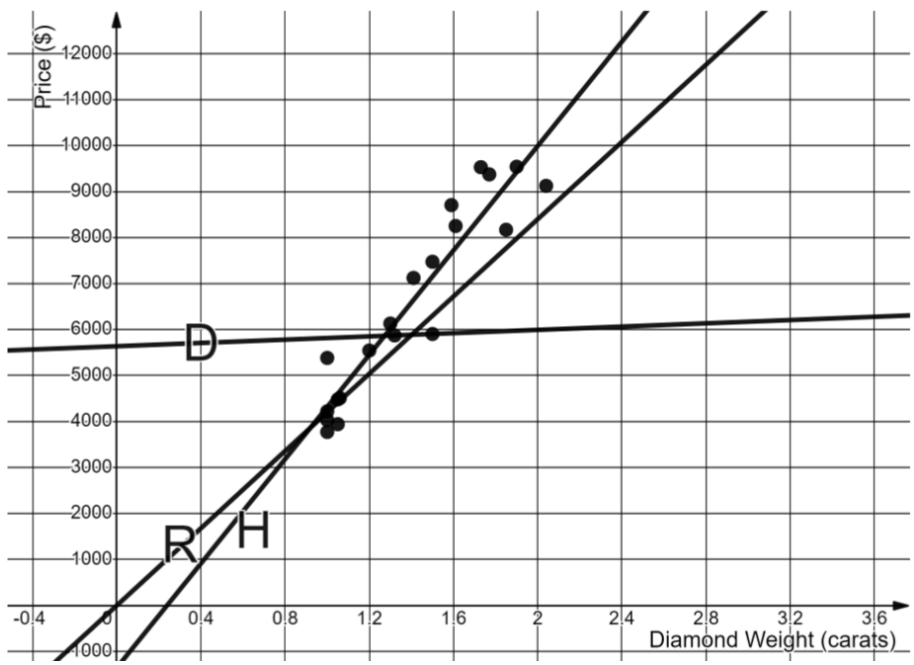
Scatter Plot F



Match each scatter plot with characteristics that are true about the data shown on the graph.

	Characteristics				
Scatter Plot	**Has a linear association**	**Has a nonlinear association**	**Positive Association**	**Negative Association**	**Outlier(s)**
A	■	□	□	■	■
B	■	□	□	■	□
C	■	□	■	□	□
D	□	■	□	□	□
E	□	■	□	□	□
F	■	□	■	□	□

3. The scatter plot shows the relationship between the weight, in carats, and the price, in dollars, of 20 different diamonds.



Complete the statement.

Line ○ D ○ H ○ R is the line of fit because

○ it goes through the origin.

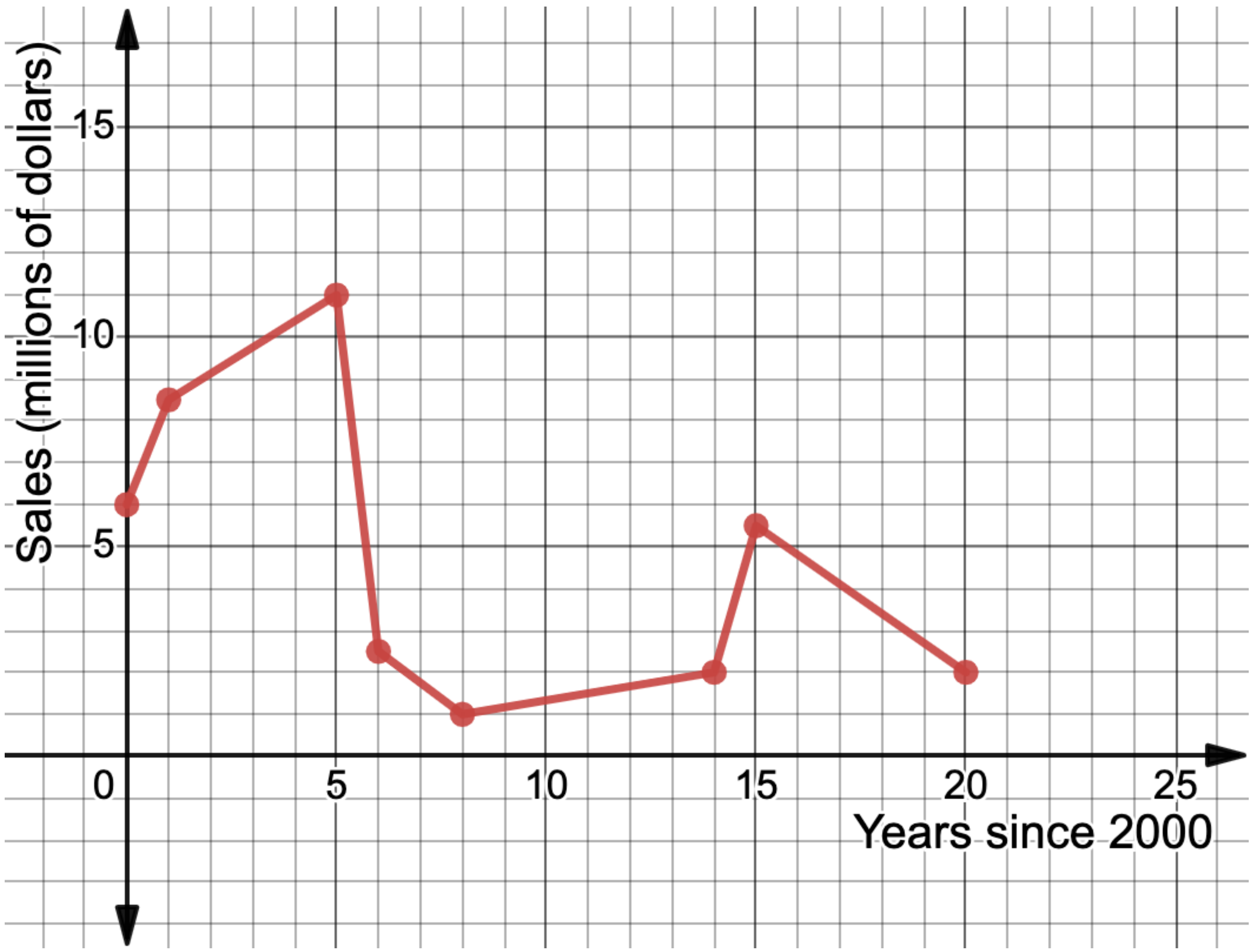
○ it goes through exactly two points and cuts the data in half.

○ approximately half of the data is above the line and half of the data is below the line.

4. DVD audio is a type of digital format for music that was first sold in 2000. The table shows sales, in millions of dollars, of audio DVDs.

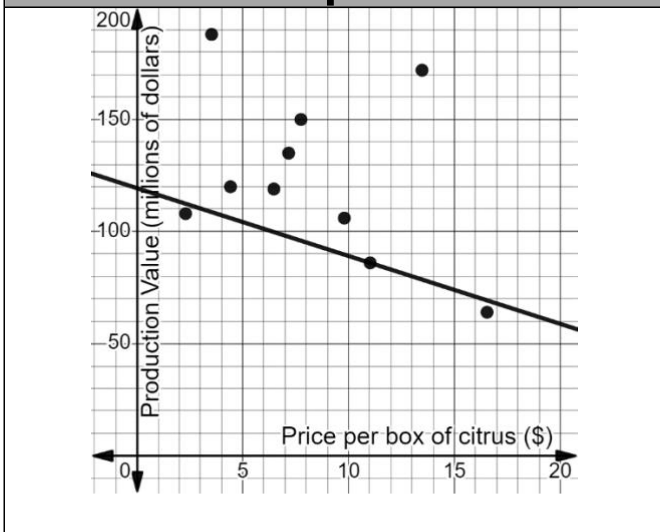
Years since 2000	0	1	5	6	8	14	15	20
Sales (millions of dollars)	6	8.5	11	2.5	1	2	5.5	2

Create a line graph for the data.

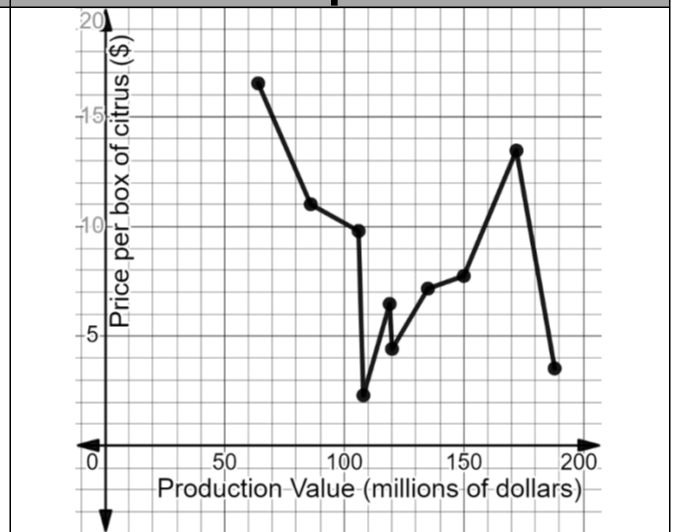


5. The Florida Department of Agriculture and Consumer Services collects data on the on-tree value of citrus, in dollars. The graphs shown relate the data for of the production value (the total amount earned), in millions of dollars, with the price per box of citrus, in dollars.

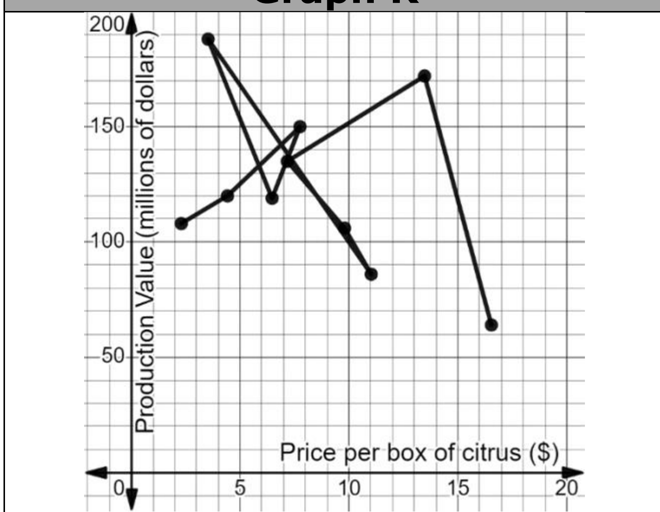
Graph K



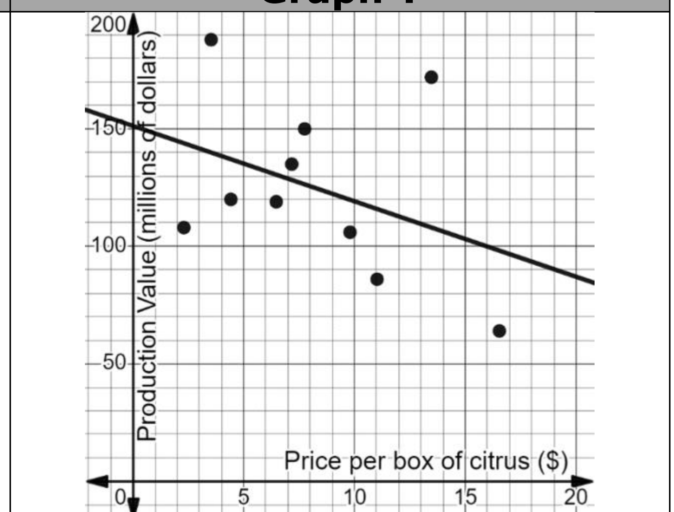
Graph N



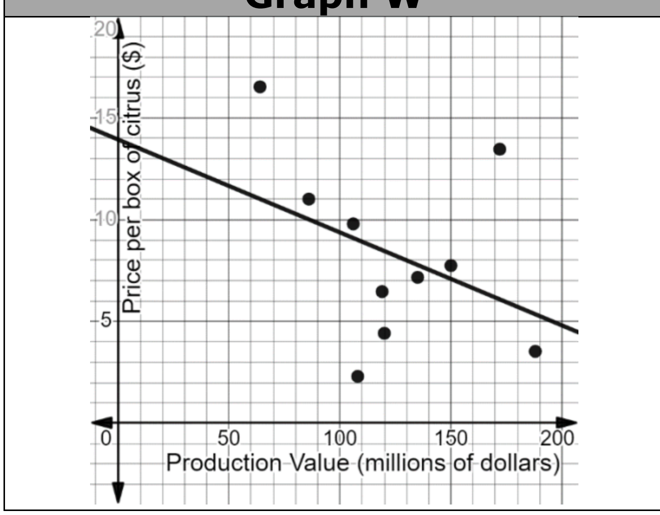
Graph R



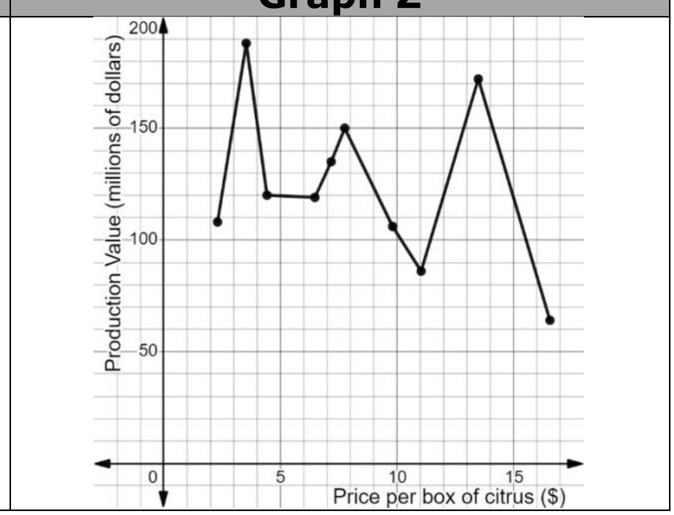
Graph T



Graph W



Graph Z



a. A statistician is interested in determining the association and overall trend of production value, in millions of dollars, for each dollar increase in the price, in dollars, of a box of citrus.

Complete the statements.

The statistician should choose to create a

- ○ line graph
- ○ scatter plot

where the

- ○ dependent
- ○ independent

variable is the price of a box of citrus, in dollars.

Graph

- ○ K
- ○ N
- ○ R
- ○ T
- ○ W
- ○ Z

is the best choice for the statistician to use.

b. An accountant is interested in representing the change in box prices, in dollars, between each set of two consecutive production values.

Complete the statements.

The accountant should choose to create a

- ○ line graph
- ○ scatter plot

where the

- ○ dependent
- ○ independent

variable is the price of a box of citrus, in dollars.

Graph

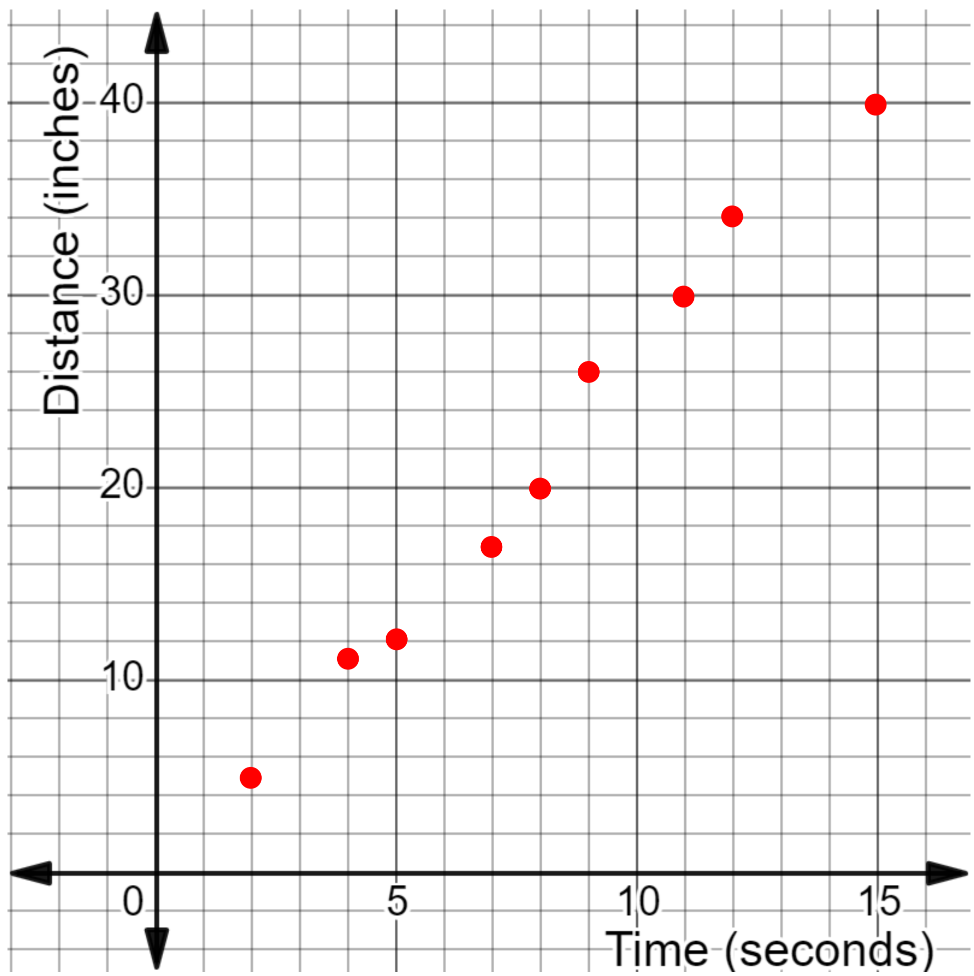
- ○ K
- ○ N
- ○ R
- ○ T
- ○ W
- ○ Z

is the best choice for the accountant to use.

6. A researcher collected data on the distance, in inches (in.), a sloth climbed on a tree and the time, in seconds (sec), the sloth took to reach that distance. Her data is shown.

Time (sec)	2	4	5	7	8	9	11	12	15
Distance (in.)	5	11	12	17	20	26	30	34	40

a. Create a scatterplot for the data.

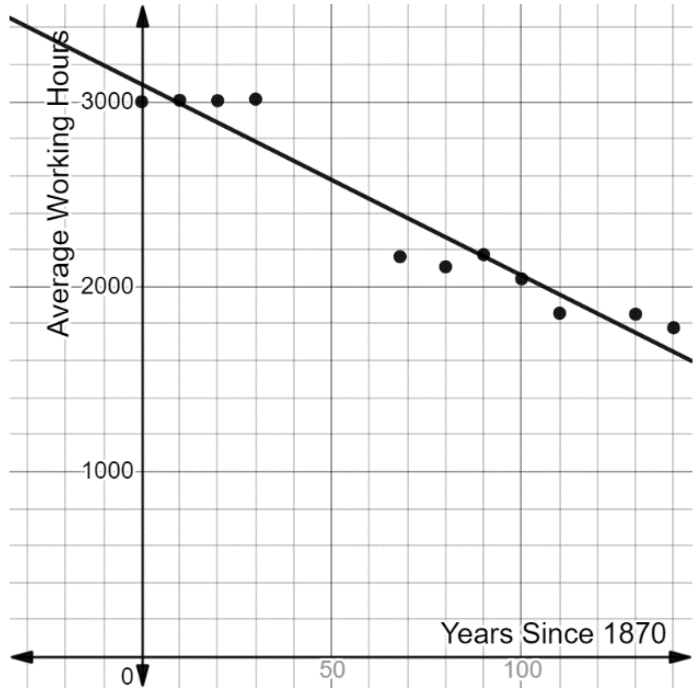


b. Complete the statement about the scatter plot.

The scatter plot shows that the data has

- ○ no association.
- ○ a nonlinear association.
- ○ a positive linear association.
- ○ a negative linear association.

7. A social scientist researched how the average working hours per worker over a full year for Italy changed since 1870. She created a scatter plot and drew the line of fit for the data, as shown.



The equation that fits the data is $y = -\frac{21}{2}x + 3094$

Complete the statements.

The slope of the equation that fits the data means that, on average, the

- ○ number of years
- ○ average working hours

- ○ increases
- ○ decreases

- ○ 2
- ○ 21
- ○ 3,094

- ○ years
- ○ working hours

for every

- ○ 2
- ○ 21
- ○ 3,094

- ○ years.
- ○ working hours.

The y-intercept means that in the year

- ○ 1870,
- ○ 3094,

the average number of working hours over a full year a worker in Italy worked was

- ○ 2.
- ○ 21.
- ○ 1,870.
- ○ 3,094.